

Soil Mechanics Analysis

Effective Stress, Bearing Capacity, and Consolidation

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Abstract

This report presents comprehensive geotechnical analysis including effective stress calculations, Mohr-Coulomb failure criteria, bearing capacity design using Terzaghi's theory, one-dimensional consolidation settlement, and slope stability assessment. Computational methods are applied to practical foundation design scenarios with parametric studies examining the influence of soil properties on engineering behavior.

1 Introduction

Soil mechanics forms the foundation of geotechnical engineering, governing the design of structures interacting with earth materials. This analysis examines five critical aspects: (1) effective stress profiles accounting for pore water pressure, (2) shear strength characterization via Mohr-Coulomb failure envelopes, (3) shallow foundation bearing capacity, (4) time-dependent consolidation settlement, and (5) slope stability under various loading conditions.

The principle of effective stress, first articulated by Terzaghi [1], remains fundamental to understanding soil behavior. Shear strength parameters (cohesion c and friction angle ϕ) control failure mechanisms [2]. Bearing capacity theory provides the basis for safe foundation design [3], while consolidation analysis predicts settlement magnitudes and time scales [4].

2 Effective Stress Distribution

The effective stress principle states that soil behavior is controlled by effective stress σ' , defined as:

$$\sigma' = \sigma - u \tag{1}$$

where σ is total stress and u is pore water pressure.

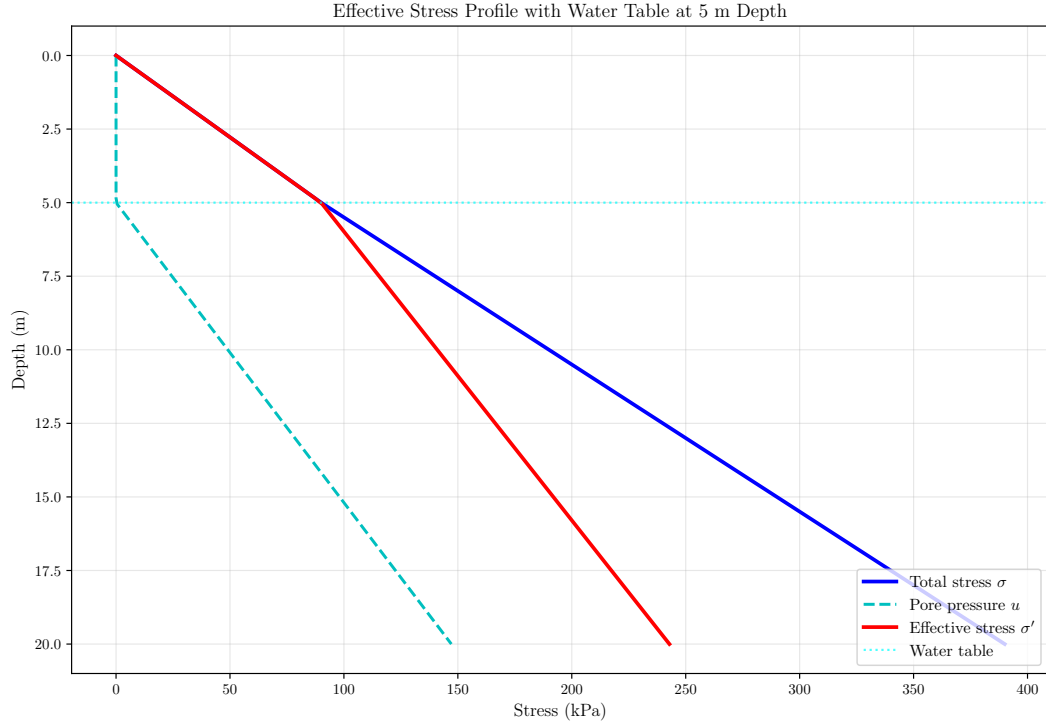


Figure 1: Effective stress distribution with depth showing the influence of the water table at 5 m depth. Above the water table, total stress equals effective stress since pore pressure is zero. Below the water table, buoyancy effects reduce effective stress significantly. At 20 m depth, total stress is 390.0 kPa, pore pressure is 147.2 kPa, and effective stress is 242.8 kPa. This principle governs compressibility, shear strength, and permeability of saturated soils.

3 Mohr-Coulomb Failure Criterion

The Mohr-Coulomb failure envelope defines shear strength τ_f as:

$$\tau_f = c + \sigma_n \tan \phi \quad (2)$$

where c is cohesion, σ_n is normal effective stress, and ϕ is the friction angle.

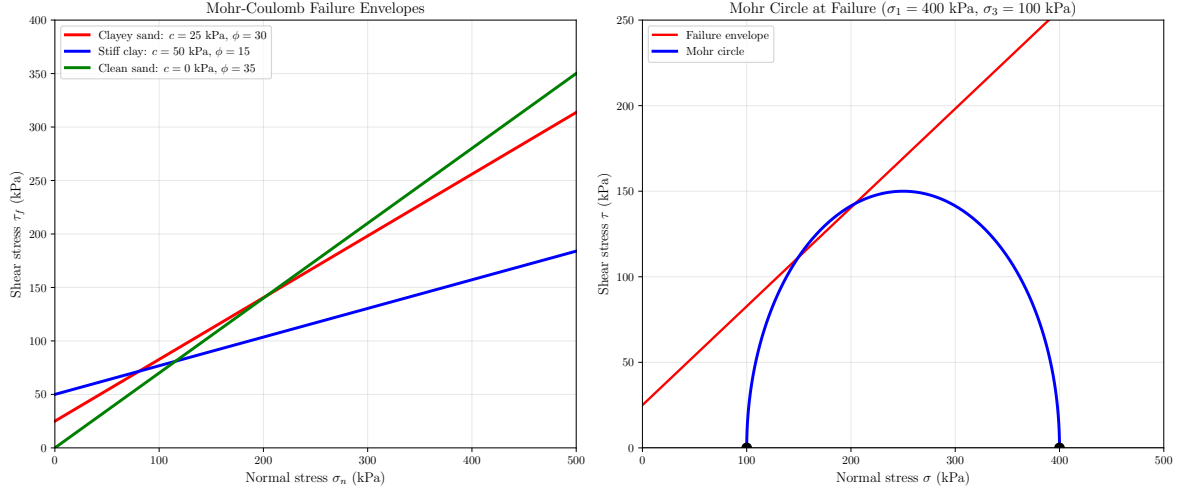


Figure 2: Left: Mohr-Coulomb failure envelopes for three soil types demonstrating how cohesion and friction angle control shear strength. Clayey sand exhibits moderate cohesion and friction. Stiff clay has high cohesion but lower friction. Clean sand has zero cohesion but highest friction angle. Right: Mohr circle representation of stress state at failure tangent to the failure envelope. The principal stresses are 400 kPa and 100 kPa, giving a deviatoric stress of 300 kPa. This graphical method is fundamental for analyzing triaxial test results and determining soil strength parameters.

4 Bearing Capacity Analysis

Terzaghi's bearing capacity equation for a shallow strip footing is:

$$q_u = cN_c + \gamma D_f N_q + \frac{1}{2} \gamma B N_\gamma \quad (3)$$

where q_u is ultimate bearing capacity, D_f is embedment depth, B is footing width, and N_c , N_q , N_γ are bearing capacity factors.

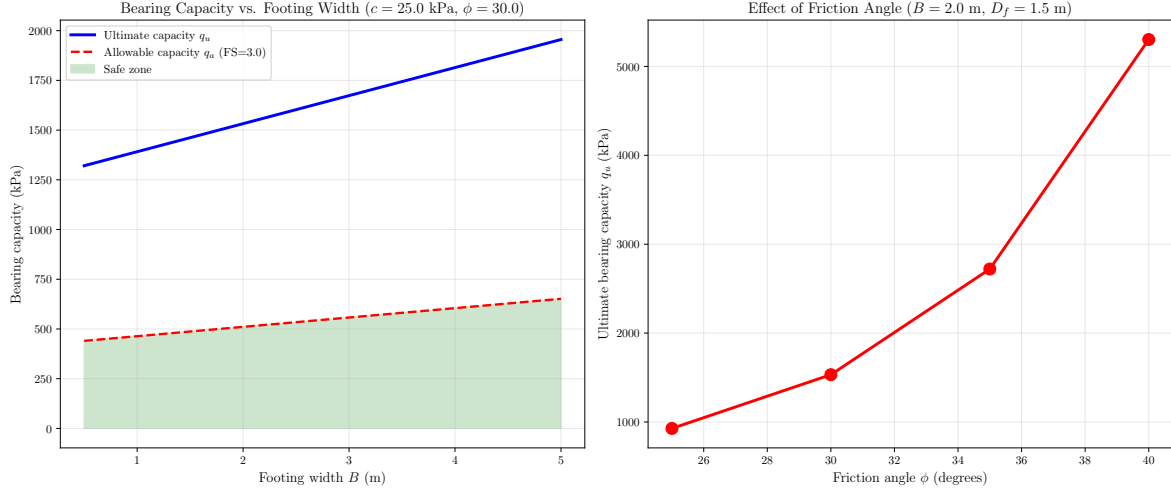


Figure 3: Left: Ultimate and allowable bearing capacity variation with footing width showing the significant influence of the size-dependent N_γ term. For a 2 m wide footing, ultimate capacity is 1532 kPa and allowable capacity is 511 kPa. Right: Bearing capacity sensitivity to friction angle demonstrating exponential increase. A 5° increase from 30° to 35° raises capacity by over 50 percent. These relationships guide foundation sizing in preliminary design. Bearing capacity factors are $N_c = 30.14$, $N_q = 18.40$, $N_\gamma = 15.67$ for $\phi = 30.0$.

5 One-Dimensional Consolidation

Terzaghi's one-dimensional consolidation theory predicts settlement and its time development. The degree of consolidation U is:

$$U = 1 - \sum_{m=0}^{\infty} \frac{2}{M^2} \exp(-M^2 T_v), \quad M = \frac{\pi}{2}(2m + 1) \quad (4)$$

where the time factor $T_v = c_v t / H_{dr}^2$, with c_v as the coefficient of consolidation and H_{dr} as the drainage path.

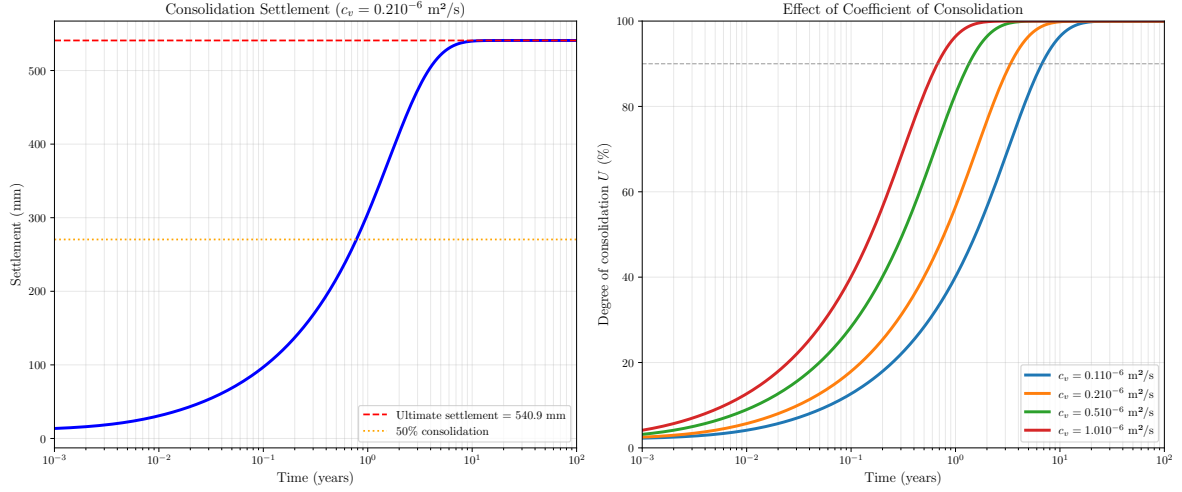


Figure 4: Left: Settlement progression over time under 100 kPa surcharge on a 10 m thick clay layer with double drainage. Ultimate settlement is 540.9 mm. Time for 50% consolidation is 0.78 years and for 90% consolidation is 3.3 years. Right: Consolidation rate sensitivity to coefficient of consolidation showing that doubling c_v halves the time required for a given degree of consolidation. Lower permeability clays ($c_v \sim 10^{-7} \text{ m}^2/\text{s}$) require decades, while silty clays ($c_v \sim 10^{-6} \text{ m}^2/\text{s}$) consolidate in months. This analysis is essential for staged construction and surcharge preloading design.

6 Slope Stability Analysis

Infinite slope stability is analyzed using limit equilibrium. The factor of safety against sliding is:

$$FS = \frac{c + \gamma z \cos^2 \beta \tan \phi}{\gamma z \sin \beta \cos \beta} \quad (5)$$

where β is the slope angle and z is the depth of failure surface.

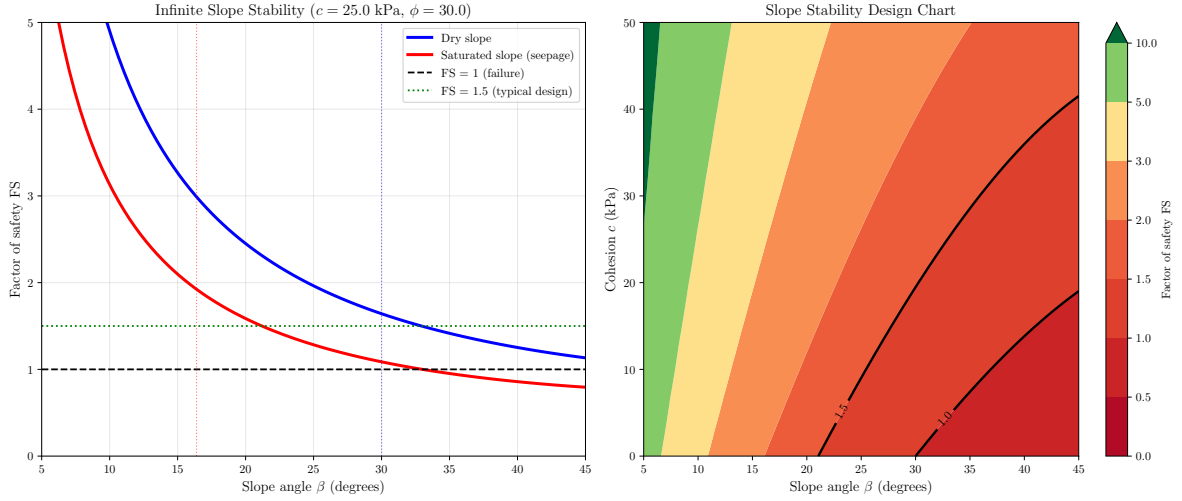


Figure 5: Left: Factor of safety versus slope angle for infinite slope showing critical importance of groundwater. For a 30° slope, dry conditions give $FS = 1.64$ while saturated conditions reduce this to $FS = 1.09$. Critical angle for cohesionless dry soil is 30° and for saturated soil is 16° . Right: Design chart showing safe (green) and unstable (red) combinations of slope angle and cohesion. Contours at $FS = 1.0$ (failure) and 1.5 (typical design minimum) provide rapid assessment. This demonstrates why cohesive soils can sustain steeper slopes and why drainage is critical for slope stability.

7 Compressibility and Settlement

Soil compressibility governs settlement under foundation loads. The compression index C_c and recompression index C_r define the stress-strain relationship.

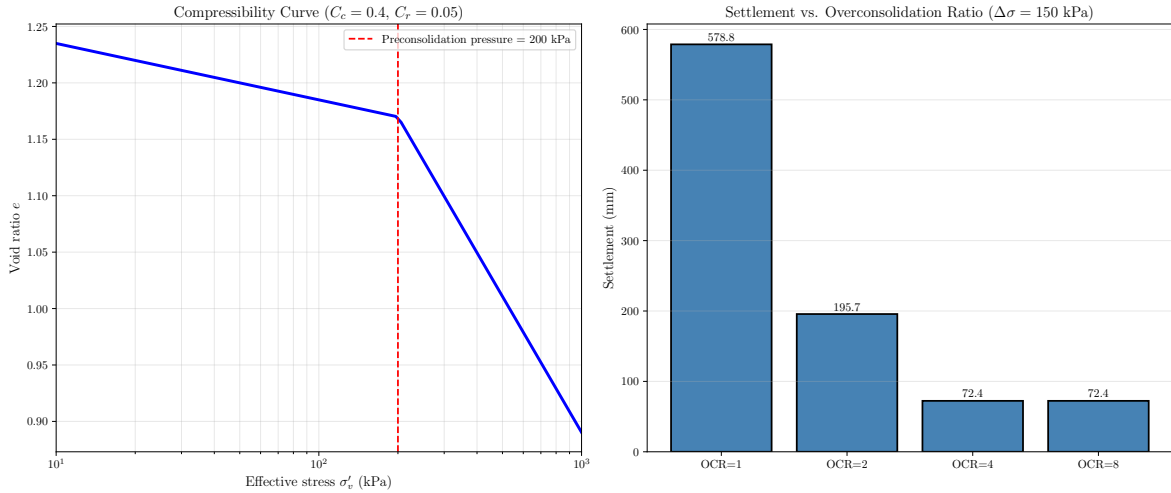


Figure 6: Left: Void ratio-effective stress relationship showing elastic recompression at low stress (slope $C_r = 0.05$) and virgin compression beyond preconsolidation pressure (slope $C_c = 0.40$). The break in slope at 200 kPa indicates stress history. Right: Settlement reduction with increasing overconsolidation ratio (OCR). Normally consolidated clay ($OCR=1$) settles 578.8 mm while heavily overconsolidated clay ($OCR=8$) settles only 72.4 mm under identical loading. This demonstrates why foundation performance depends critically on stress history, motivating preloading and surcharge techniques to reduce post-construction settlement.

8 Results Summary

The computational analysis yields the following key engineering parameters:

Table 1: Summary of Computed Geotechnical Parameters

Parameter	Value
Ultimate bearing capacity (B=2m)	1532 kPa
Allowable bearing capacity (FS=3)	511 kPa
Ultimate consolidation settlement	540.9 mm
Time for 50% consolidation	0.78 years
Time for 90% consolidation	3.3 years
Bearing capacity factor N_c	30.14
Bearing capacity factor N_q	18.40
Bearing capacity factor N_γ	15.67
Slope FS (30°, dry)	1.64
Slope FS (30°, saturated)	1.09

9 Conclusions

This comprehensive analysis demonstrates computational methods for five fundamental aspects of soil mechanics. Effective stress analysis reveals that pore water pressure significantly reduces effective stress below the water table, at 20 m depth reducing effective stress by 37.7% compared to total stress. Mohr-Coulomb failure analysis shows shear strength is controlled by both cohesion ($c = 25.0$ kPa) and friction angle ($\phi = 30.0$), with clean sands relying entirely on friction while clays exhibit significant cohesion.

Bearing capacity calculations using Terzaghi's theory yield ultimate capacity $q_u = 1532$ kPa for a 2 m wide footing, giving allowable capacity $q_a = 511$ kPa with factor of safety FS = 3.0. Parametric studies show bearing capacity increases exponentially with friction angle, emphasizing the importance of accurate shear strength characterization.

One-dimensional consolidation analysis predicts ultimate settlement of 540.9 mm under 100 kPa surcharge, with 90% consolidation requiring 3.3 years. This time-dependent behavior governs staged construction schedules and surcharge design. Slope stability analysis demonstrates the critical importance of groundwater, with saturation reducing factor of safety from 1.64 to 1.09 for a 30° slope, explaining the prevalence of slope failures during heavy rainfall.

Compressibility analysis shows that overconsolidation ratio dramatically affects settlement, with OCR = 8 reducing settlement by 88% compared to normally consolidated soil. This stress history effect is exploited in ground improvement through preloading. The integrated computational framework presented here provides quantitative design tools for foundation engineering, earthwork stability, and settlement prediction, forming the basis for safe and economical geotechnical design.

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